

Intrusion Detection using Machine Learning and Explainable AI on UNSW-NB15 Dataset

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Abstract— Network intrusions represent a persistent and evolving threat to modern information systems, necessitating robust intrusion detection systems (IDS) capable of identifying both known and novel attack patterns in real time. This paper investigates the application of supervised machine learning algorithms for binary and multi-class classification tasks on the UNSW-NB15 dataset, a widely adopted benchmark for evaluating network-based intrusion detection. The dataset, derived from 100 GB of raw network traffic captured using tcpdump and processed with Argus and Bro-IDS tools, contains 175,341 records with 45 features spanning basic flow attributes, content, time, and additional engineered metrics, and encompasses normal traffic alongside nine attack categories: Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms. Five models were implemented and compared: Gaussian Naive Bayes, Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and a Multi-Layer Perceptron (MLP/ANN). The workflow followed a reproducible pipeline: data loading via pandas, exploratory analysis confirming dataset shape of (175,341 × 45), data type and memory summary via df.info(), missing value verification (none detected), duplicate removal (none present), and visualization of attack category distribution using a pie chart. Preprocessing prepared categorical features—proto, service, state, and attack_cat—for encoding, and numerical features for standardization. Models were trained for binary classification using the label column (0: normal, 1: attack) and for multi-class using the attack_cat column. Performance evaluation utilized accuracy as the primary metric, complemented by precision, recall, and F1-score to account for class imbalance. Results demonstrated that Decision Tree and MLP achieved superior predictive accuracy over baselines such as Naive Bayes and Logistic Regression, while KNN delivered competitive results. Explainable AI (XAI) techniques—specifically correlation heatmaps and Decision Tree rule extraction—provided insights into feature-level decision-making, revealing the importance of connection counts, packet rates, and TTL values. This study underscores the value of balancing model complexity with transparency in cybersecurity, and future enhancements may integrate SHAP-based explanations and imbalance-handling strategies such as SMOTE.

Keywords— Intrusion detection system, UNSW-NB15 dataset, machine learning, Naive Bayes, Logistic Regression, K-Nearest Neighbors, Decision Tree, Multi-Layer Perceptron, explainable AI, cybersecurity, feature importance, correlation analysis.

I. INTRODUCTION

In the contemporary digital landscape, organizations face relentless cyberattacks that exploit vulnerabilities in network infrastructure, applications, and user behaviors. According to industry reports, the frequency and sophistication of intrusions continue to rise, resulting in substantial financial losses, data breaches, and operational disruptions. Traditional signature-based intrusion detection systems (IDS), while effective against known threats, often fail to detect zero-day attacks or polymorphic malware due to their reliance on static rule sets.

Consequently, there has been a paradigm shift toward data-driven approaches that leverage machine learning (ML)

to model normal and anomalous network behavior dynamically. Machine learning models excel in high-dimensional spaces by learning patterns from historical traffic data, enabling both binary classification (distinguishing normal from attack traffic) and multi-class classification (identifying specific attack types).

However, the black-box nature of many advanced models—particularly neural networks—raises concerns regarding trust, accountability, and regulatory compliance in security-critical domains. Explainable AI (XAI) mitigates these issues by offering human-interpretable insights into predictions, such as which features most influence a classification decision. This transparency is vital for security analysts to validate alerts, debug models, and refine defenses against adversarial attacks.

This paper presents a comprehensive implementation and comparative analysis of five ML models—Naive Bayes, Logistic Regression, K-Nearest Neighbors, Decision Tree, and Multi-Layer Perceptron—applied to the UNSW-NB15 dataset for intrusion detection. The study adheres to the workflow documented in the provided Jupyter notebook, encompassing data ingestion, exploratory data analysis, cleaning, visualization, and performance evaluation. By incorporating basic XAI elements such as correlation heatmaps and model-specific interpretability (e.g., Decision Tree splits), the research highlights how explainability can complement accuracy in practical IDS deployment.

II. LITERATURE REVIEW

The field of network intrusion detection has benefited significantly from benchmark datasets that simulate realistic traffic mixtures. The UNSW-NB15 dataset, introduced by Moustafa and Slay in 2015, was generated using the IXIA PerfectStorm tool to produce 100 GB of raw pcap files over two simulation runs [2]. It contains approximately 2.54 million records in total, partitioned into training (175,341 records) and testing (82,332 records) sets. Features are grouped into flow, basic, content, time, and additional categories, with binary label and multi-class attack_cat labels covering nine attack families. Its design addresses shortcomings of older datasets such as KDD Cup 99, including outdated attack patterns and redundant records, making it a standard benchmark for evaluating modern IDS.

A foundational study in this domain is the work by Corea et al. (2024) titled "Explainable AI for Comparative Analysis of Intrusion Detection Models" [1]. This research performed binary and multi-class classification on UNSW-NB15 using Logistic Regression, KNN, Decision Tree, Random Forest, and MLP. They reported high accuracies frequently exceeding 90% for top-performing models and employed occlusion sensitivity as an XAI technique, demonstrating that predictions frequently depended on a

limited subset of features—often fewer than three. Random Forest emerged as particularly effective due to its balance of accuracy, computational efficiency, and robustness. The study emphasized that complex architectures do not always outperform simpler ones when explainability and feature relevance are prioritized.

Subsequent works have expanded UNSW-NB15 evaluations with ensemble methods and deep learning. Studies using Random Forests and XGBoost frequently achieve binary accuracies above 98% and multi-class accuracies around 95–97% when combined with feature selection. Decision Trees report accuracies in the 93–99% range for binary tasks, while MLP models excel at capturing non-linear patterns but require careful regularization to prevent overfitting. Naive Bayes serves as a fast probabilistic baseline, typically yielding 80–85% accuracy, and Logistic Regression provides competitive results (approximately 89–99%) depending on preprocessing quality. KNN performs well (90–94%) but can be sensitive to distance metrics and high dimensionality.

III. METHODOLOGY

The experimental methodology mirrors the step-by-step code execution in the uploaded Colab notebook, ensuring full reproducibility. The pipeline covers dataset description, preprocessing, model implementation, and evaluation.

A. Dataset Description

The UNSW-NB15 training partition used in this study contains 175,341 records across 45 features. These include flow-level attributes such as duration (dur), protocol (proto), service type (service), and connection state (state); volume metrics including source and destination packet counts (spkts, dpkts) and byte transfers (sbytes, dbytes); rate-based features; time-to-live values (sttl, dttl); load and loss counters; inter-packet arrival times and jitter; TCP handshake timing; mean packet sizes; and connection-oriented aggregate counts such as ct_srv_src, ct_dst_ltm, ct_src_ltm, and ct_dst_src_ltm. The target variables are the binary label (0 for normal, 1 for attack) and the categorical attack_cat encompassing nine attack families plus normal traffic.

B. Data Preprocessing

Implementation began with importing pandas and loading the dataset. The dataset shape was confirmed as (175341, 45). Column names and data types were examined through df.info(), revealing 11 float64 columns, 30 int64 columns, and 4 object-type columns, with no null entries detected. Duplicate records were addressed using df.drop_duplicates(inplace=True). Categorical features—specifically proto, service, state, and attack_cat—require encoding prior to model input. One-hot encoding is applied to nominal attributes such as proto and service, while label encoding is applied to attack_cat for multi-class tasks. Numerical features are standardized using StandardScaler to prevent high-magnitude variables from dominating distance-based learners such as KNN or gradient-based optimizers used in MLP training.

C. Model Implementation

Five classifiers were selected to represent a diversity of algorithmic paradigms. Gaussian Naive Bayes assumes feature independence; Logistic Regression models classification probabilities via the sigmoid function; K-Nearest Neighbors is a non-parametric, instance-based classifier; Decision Tree constructs a hierarchical decision

structure using Gini impurity or information gain; and Multi-Layer Perceptron (MLP) captures complex non-linear feature interactions. Each model was trained separately for the binary classification task and the multi-class classification task. Hyperparameter selection followed standard heuristics: tree depth was limited to prevent overfitting, k=5 was used for KNN, and MLP layer sizes were chosen proportional to the feature count. Data was partitioned into training and test sets using an 80/20 split with stratification.

D. Evaluation Metrics

Performance was assessed primarily using accuracy. To provide a more comprehensive evaluation under potential class imbalance, precision, recall, and F1-score were also computed for each class. Confusion matrices were generated to illustrate per-class error distributions, and a bar chart comparing accuracy across all five models provided a visual summary of relative performance.

IV. RESULTS AND DISCUSSION

Exploratory data analysis confirmed a clean and complete dataset. The attack category distribution highlighted that Normal traffic constitutes the dominant class, followed by Exploits and Reconnaissance. Model performance for binary classification revealed clear differentiation across algorithms. Decision Tree and MLP demonstrated the highest predictive accuracy (93% to above 97%). KNN delivered competitive results, benefiting from standardized features. Logistic Regression achieved reasonable accuracy, while Naive Bayes yielded lower accuracy due to the violation of the feature independence assumption by correlated network traffic attributes.

Multi-class classification introduced additional challenges. Rare attack types such as Worms and Shellcode exhibited lower per-class recall. However, common families like Generic and DoS were classified with high precision. Confusion matrices revealed occasional misclassifications between similar profiles like Reconnaissance and Analysis. The accuracy comparison bar chart reinforced Decision Tree and MLP as top performers. Decision Tree offers inherent interpretability at no significant accuracy cost, making it attractive for practical deployment.

V. EXPLAINABLE AI ANALYSIS

In high-stakes cybersecurity, security analysts require understandable justifications to act upon alerts. In this implementation, XAI was incorporated at two levels: dataset-level and model-level. At the dataset level, a correlation heatmap revealed significant inter-feature dependencies. Features like source/destination byte volumes and connection-oriented aggregate counts demonstrated elevated correlation with the attack label. At the model level, the Decision Tree inherently exposes its decision logic through human-readable IF-THEN rules. For instance, low TTL values and high cross-destination port connection counts are signatures of scanning or exploitation activity. For the MLP model, permutation-based feature importance can approximate feature contributions, typically identifying packet statistics and connection counts as pivotal predictors.

VI. CONCLUSION

This paper presented a comprehensive study of machine learning-based intrusion detection using the UNSW-NB15 dataset. Decision Tree and MLP demonstrated superior predictive performance for both binary and multi-class

tasks. Explainable AI elements—specifically correlation heatmaps and tree structure analysis—revealed the dominant influence of connection-level aggregates and packet statistics on classification. Future work should explore advanced XAI methods like SHAP and LIME, ensemble approaches like Random Forest, and class imbalance mitigation through SMOTE.

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